

```
mysql> CREATE DATABASE HospitalDB;
Query OK, 1 row affected (0.03 sec)

mysql> USE HospitalDB;
Database changed
mysql> CREATE TABLE physician (
  ->     employeeid INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     position VARCHAR(100),
  ->     ssn VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.12 sec)

mysql>
mysql> CREATE TABLE department (
  ->     departmentid INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     head INT
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT d.name AS Department,
  ->     p.name AS Head_Physician
  -> FROM department d
  -> JOIN physician p
  -> ON d.head = p.employeeid;
Empty set (0.02 sec)

mysql> SELECT *
  -> FROM physician
  -> WHERE position='Surgical Attending Physician';
Empty set (0.00 sec)
```

```
mysql> SELECT *
-> FROM physician
-> WHERE name LIKE 'John%';
Empty set (0.02 sec)
```

```
mysql> SELECT COUNT(*) AS Total_Physicians
-> FROM physician;
+-----+
| Total_Physicians |
+-----+
|                0 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(DISTINCT position)
-> FROM physician;
+-----+
| COUNT(DISTINCT position) |
+-----+
|                0 |
+-----+
1 row in set (0.03 sec)
```

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position;
Empty set (0.00 sec)
```

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) > 1;
Empty set (0.00 sec)
```

```
mysql> SELECT *
-> FROM physician
-> ORDER BY name ASC;
Empty set (0.00 sec)

mysql> SELECT *
-> FROM physician
-> ORDER BY employeeid DESC;
Empty set (0.00 sec)

mysql> SELECT *
-> FROM physician
-> WHERE employeeid IN
-> (
->     SELECT head
->     FROM department
-> );
Empty set (0.02 sec)

mysql> SELECT *
-> FROM physician
-> WHERE employeeid NOT IN
-> (
->     SELECT head
->     FROM department
-> );
Empty set (0.00 sec)

mysql> SELECT position,
->     COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> ORDER BY Total_Employees DESC;
Empty set (0.00 sec)
```

```

mysql> SELECT *
-> FROM physician
-> WHERE position LIKE '%Attending%';
Empty set (0.00 sec)

mysql> SELECT p.name
-> FROM physician p
-> JOIN department d
-> ON p.employeeid = d.head
-> WHERE d.name = 'Surgery';
Empty set (0.00 sec)

mysql> SELECT d.name,
->      p.name
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid
-> ORDER BY d.name;
Empty set (0.00 sec)

mysql> SELECT *
-> FROM physician
-> WHERE position NOT LIKE '%Surgical%';
Empty set (0.02 sec)

mysql> SELECT position,
->      COUNT(*) AS Total
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) >= 2;
Empty set (0.00 sec)

```

```

mysql> SELECT *
-> FROM department d
-> WHERE EXISTS
-> (
->      SELECT 1
->      FROM physician p
->      WHERE p.employeeid = d.head
-> );
Empty set (0.00 sec)

mysql> SELECT MAX(employeeid)
-> FROM physician;
+-----+
| MAX(employeeid) |
+-----+
|                |
+-----+
1 row in set (0.02 sec)

mysql> SELECT MIN(employeeid)
-> FROM physician;
+-----+
| MIN(employeeid) |
+-----+
|                |
+-----+
1 row in set (0.00 sec)

```

```
mysql> USE HospitalDB;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_hospitaldb |
+-----+
| department            |
| physician              |
+-----+
2 rows in set (0.02 sec)

mysql> SELECT * FROM physician;
Empty set (0.00 sec)

mysql> SELECT * FROM department;
Empty set (0.00 sec)

mysql> SELECT d.name AS Department,
->         p.name AS Head_Physician
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid;
Empty set (0.00 sec)

mysql> SELECT COUNT(*) FROM physician;
+-----+
| COUNT(*) |
+-----+
|         0 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO physician VALUES
-> (1,'John Dorian','Staff Internist','111111111'),
-> (2,'Elliot Reid','Attending Physician','222222222'),
-> (3,'Christopher Turk','Surgical Attending Physician','333333333'),
-> (4,'Percival Cox','Senior Attending Physician','444444444'),
-> (5,'Bob Kelso','Head Chief of Medicine','555555555'),
-> (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
-> (7,'John Wen','Surgical Attending Physician','777777777'),
-> (8,'Keith Dudemeister','MD Resident','888888888'),
-> (9,'Molly Clock','Attending Psychiatrist','999999999');
Query OK, 9 rows affected (0.02 sec)
Records: 9 Duplicates: 0 Warnings: 0

mysql> INSERT INTO department VALUES
-> (1,'General Medicine',4),
-> (2,'Surgery',7),
-> (3,'Psychiatry',9);
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM physician;
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
4	Percival Cox	Senior Attending Physician	444444444
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM department;
```

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT d.name AS Department,
->         p.name AS Head_Physician
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid;
```

Department	Head_Physician
General Medicine	Percival Cox
Surgery	John Wen
Psychiatry	Molly Clock

```
3 rows in set (0.00 sec)
```

```
mysql>
```